

CloudX Demo Scripts

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Prep/Prerequisite

Must have Ollama installed on laptop and configured.

Must have Powershell and use in Admin role.

Ensure all environment is set in PATH for Postgres, Ollama, Flyway

I've configured mine to use the postgres user, but you could change this for enhanced security.

Ensure Postgres is running and has a demo database. If needed, set `$env:PGPASSWORD`.

Unzip project; ensure psql and **flyway are on PATH** (or set FYW_EXE in scripts).

```
createdb -h localhost -U postgres demo
```

```
$env:PGPASSWORD = "use password"
```

The `pg_stat_statements` extension must be installed/configured

After downloading the scripts, you may have to unblock the scripts in each of the file properties due to security in Windows.

Pipeline (Governance/Migrate/Smoke/Observe)

cd to the scripts directory where files were extracted:

Run the first script:

```
.\scripts\run.ps1
```

Governance: `verify-hash.ps1` checks SHA-256 of migrations.

Migrate: Flyway applies `sql/V1..V4` (GitOps-style versioned migrations).

Smoke: Confirms objects exist and RLS is enabled on invoices.

Observe: Stage timings; tries `pg_stat_statements` (skips if not permitted).

Review the output screen

*Go over the ai code review md file to look for recommendations on how code can be improved.

2) Seed & Prove RLS

`.\scripts\seed.ps1`

This is the full code of what will be done in the Canary test if we were to do it in reality... 😊

```
psql -h ${HostName} -p $Port -U $User -d $Database -c "ALTER TABLE public.customers ENABLE ROW LEVEL SECURITY; ALTER TABLE public.customers FORCE ROW LEVEL SECURITY;"
```

```
psql -h ${HostName} -p $Port -U $User -d $Database -c "ALTER TABLE public.invoices ENABLE ROW LEVEL SECURITY; ALTER TABLE public.invoices FORCE ROW LEVEL SECURITY;"
```

```
psql -h ${HostName} -p $Port -U $User -d $Database -c "CREATE POLICY rls_customers_tenant ON public.customers USING (tenant_id = current_setting('app.tenant_id', true));"
```

```
psql -h ${HostName} -p $Port -U $User -d $Database -c "CREATE POLICY rls_invoices_tenant ON public.invoices USING (tenant_id = current_setting('app.tenant_id', true));"
```

- Policy uses `current_setting('app.tenant_id')`; helper `set_tenant()` sets per session.
- Least-privilege role `app_readonly` granted `SELECT` only.

Progressive Delivery via Feature Flag (Canary)

`.\scripts\toggle-canary.ps1 -Enable:$true`

Script will enable Canary

Set the RLS

Check the views

Report back with a 1 if valid

`.\scripts\toggle-canary.ps1 -Enable:$false`

Will disable Canary once complete

- V3 adds v_invoices_current with 5% discount when demo.canary_enabled=on.
- V4 reverts via roll-forward (safe rollback pattern).

Optional CI & GitOps (2 min)

CI: review .github/workflows/ci.yml (Postgres service integrity migrate smoke canary).

```
act -W .github/workflows/ci.yml -j ci --secret PGPASSWORD=postgres
```

GitOps (preview only):

```
.\ops\scripts\render-kustomize.ps1 -Overlay stable
```

```
.\ops\scripts\render-kustomize.ps1 -Overlay canary
```

Cleanup

From psql as the postgres user:

```
drop database demo;
```

Or from the command line:

```
dropdb -h localhost -U postgres demo
```

Troubleshooting

Auth: set \$env:PGPASSWORD or pass -Password to scripts.

Flyway not found: set FYW_EXE in scripts\run.ps1.

pg_stat_statements warning: needs superuser; safe to ignore in demo.

Custom port: pass -Port 5432 to scripts.

“ps1 is not digitally signed”: Right click on each file to run these powershell scripts in your own secluded test environment and choose to “unblock” from the general properties:

Created: Thursday, August 28, 2025, 5:19:28 PM

Modified: Thursday, August 28, 2025, 1:36:12 PM

Accessed: Today, August 28, 2025, 2 hours ago

Attributes: ☐ Read-only ☐ Hidden

Security: This file came from another computer and might be blocked to help protect this computer. ☐ Unblock

File Map

- sql/V1..V4 schema, RLS, feature-flag view, revert
- scripts/verify-hash.ps1 integrity guardrails
- scripts/ai-review.ps1 optional Ollama LLM review
- .github/workflows/ci.yml CI mirrors local pipeline
- app/k8s/overlays/* stable vs canary (Kustomize)